

Overcoming Discount Window Stigma: An Experimental Investigation

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A core responsibility of the Federal Reserve is to ensure financial stability by acting as the “lender of last resort” through its discount window (DW). Historically, however, the DW has not been effective because its usage is stigmatized. In this paper, we develop a coordination game with adverse selection, and we test in the lab policies that have been proposed to mitigate DW stigma. We find that lowering the DW cost and making DW borrowing difficult to detect are ineffective, but regular random DW borrowing can overcome DW stigma. Implications for other forms of stigma in finance are discussed. (*JEL* E58, G01, C92)

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In August 2007, ... banks were reluctant to rely on discount window credit to address their funding needs. The banks’ concern was that their recourse to the discount window, if it became known, might lead market participants to infer weakness—the so-called stigma problem.

—Fed Chairman Bernanke (2009)

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The discount window is the Fed's main tool for distributing money to the banking system in emergencies when bank-to-bank lending is not functioning, which is precisely the situation that prevailed in late 2007. But because banks were very reluctant to use the window, that key tool was broken.

—Fed Governor Duke (2010)

[Banks] deliberately did not ask for the liquidity they needed for fear of damaging their reputation—the “stigma” problem ... I do not think we were conscious of this before the crisis started and I do not think central banks have a convincing answer to it ... This is, I think, still a challenge in how to manage the process of central bank provision of liquidity support. This is one of the big intellectual issues that has not been fully resolved.

—Bank of England Governor King (2016a)

This paper is concerned with “discount window” (DW) stigma, defined as the reluctance to access the central bank backstop lending facility even for benign reasons, out of concerns that it could be interpreted as a sign of financial weakness. As explained below, DW stigma is a threat to financial stability and an impediment to the mission of central banks. However, little is known about DW stigma, in large part because of the absence of data. The object of this paper is to understand better DW stigma and to explore how the problem could be mitigated. To do so, we model emergency funding as a coordination game with adverse selection. The game has two equilibria, with and without DW stigma, and we test in the lab which equilibrium emerges under different policy environments. To the best of our knowledge, this is the first paper using experimental methods to study DW stigma and more generally the provision of emergency liquidity to the financial system.

In principle, solvent but illiquid banks should obtain funding from private counterparties in the interbank market. Various frictions, however, can impair the interbank market. In particular, asymmetric information can prevent lenders from distinguishing illiquid from insolvent borrowers, especially when markets are stressed. As a result, even solvent institutions can fail to secure funding on the interbank market, in which case they have to resort to costly alternatives, such as fire sales of assets, borrowing at high rates through different channels, or failing to (re)pay a counterparty. As illustrated by the rapid demise of Northern Rock in 2007, these actions can trigger a bank run, insolvency and ultimately, the failure of an illiquid yet *ex ante* solvent institution (Shin 2009). The inability to secure private funding can also have negative externalities and social costs. In particular, spillovers from fire sales can result in insolvency cascades (Shleifer and Vishny 2011). The 2008 failure of Lehman Brothers

illustrates how exposure to a failing bank can lead to contagion in a highly interconnected financial system (Caballero and Simsek 2013). Ultimately, as the events in the second half of 2007 demonstrate, adverse selection in the interbank market can start or deepen a financial crisis, leading to a contraction in economic activity (Geithner 2014).

To address market failures in the interbank market and prevent the negative externalities they can generate, central banks typically assume “lender of last resort” (LoLR) responsibilities. The objective of the LoLR is to avoid unnecessary and socially costly failures by providing liquidity support when private alternatives are not available or prohibitively expensive. In the United States, the Federal Reserve (the Fed) has been operating as a LoLR through its “discount window” for more than a century. As explained in Internet Appendix 1, the DW is a standing facility that provides sound depository institutions backup funding against a broad range of collateral at a premium above the federal funds target rate (or range). Because it aims to address any liquidity problems before they have systemic consequences, the DW is the Fed’s first line of defense against a financial crisis.

Historically, however, the DW has been scarcely used, even when banks faced acute liquidity shortages. Most notably, the DW was scarcely accessed at the onset of the 2007 financial crisis despite several measures enacted to encourage DW borrowing (see Internet Appendix 1). As illustrated by Chairman Bernanke’s quote above, this lack of DW borrowing is generally attributed to stigma. The argument goes as follows: solvent institutions that cannot meet their funding needs in private markets for benign reasons refrain from borrowing at the DW out of concern that, if detected, they might be perceived as insolvent by market participants. A second form of stigma is sometimes mentioned whereby banks may be concerned that DW borrowing sends a negative signal to the Fed, which acts both as a LoLR and as a regulator. Following the literature, we focus on DW stigma with respect to market participants.

DW stigma is a first-order concern for central banks because it can affect two of their core responsibilities: acting as a LoLR and implementing monetary policy. A central bank can only fulfill its LoLR obligations if financial institutions are willing to access the DW. As illustrated by Governor Duke’s quote above, stigma inhibited DW borrowing in the second half of 2007, which prevented the Fed from containing the emerging crisis. In normal times, DW stigma can also impair the implementation of monetary policy. To control market rates, the Fed introduced a “corridor system” in 2008. In principle, the floor of the corridor should be the interest paid on excess reserves, and the ceiling should be the DW rate. In that case, there should be no incentives to borrow above the ceiling or lend below the floor.¹ When stigma drives financial institutions to borrow above the DW rate, however, the ceiling becomes

¹ In the current high reserves environment, the Fed’s corridor system has effectively become a “floor system.” The Bank of England (BoE), the Bank of Japan, and the European Central Bank also rely on a corridor system.

“leaky” and the Fed loses its ability to control market rate fluctuations (Kahn 2010). It is well documented that stigma was a major concern in the design of the international policy response to the 2007 crisis (Geithner 2014; Winters 2012). Since then, DW stigma has been identified as a significant threat to financial stability (Duke 2010), and central banks either have reformed their DW with the explicit objective of mitigating stigma (e.g., the BoE in 2015) or are contemplating such reforms (e.g., the Fed).

Despite its relevance, DW stigma is not well understood, as illustrated by Governor King’s quote above. As explained in Section 1, only a few theoretical models have been developed in which stigma emerges in equilibrium, while evidence is scarce. So far, the literature has focused on documenting and rationalizing DW stigma, while the question of how to mitigate DW stigma has been left essentially unaddressed. As noted by Vossmeier (2019), the major challenge for applied economists is the lack of data: if financial institutions avoid the DW regardless of the policy environment, then analysts do not have data to study the problem.

As argued by, for example, Falk and Heckman (2009), experimental methods offer two key advantages in this context. First, data can be generated easily and cost effectively in the lab; second, the policies that regulate the DW can be modified exogenously in a controlled environment, thereby enabling causal inference. Such lab experiments have proved particularly useful for designing new markets and policies (for reviews, see Normann and Ricciuti 2009 or Roth 2016). Further, controlled lab experiments are used increasingly to study banks’ behavior in the interbank market (Davis et al. 2019; Duffy, Karadimitropoulos, and Parravano 2019) or in credit markets (Brown and Zehnder 2010; Bosch-Rosa 2018).² In particular, Dufwenberg (2015) emphasizes the role lab experiments can play in helping understand banking crises and testing policy responses. Nevertheless, one may question whether subjects interacting in artificial lab environments can capture actual banks’ behavior. Although recent evidence suggests that treatment effects generally carry over from typical lab subjects to professionals (Fréchette 2016; Snowberg and Yariv 2018), we need to be cognizant of external validity concerns. So, while this paper provides original insights on how to address DW stigma, it can only be considered a first step toward a complete understanding of the issue.

To study DW stigma in the lab, we develop a relatively simple coordination game with adverse selection. There are two types of players, banks and investors. The bank receives privately a solvency shock and an *end-of-day* liquidity shock, which can be addressed only at the DW or with an alternative.

² There is also a long history of lab subjects playing the role of large firms in, for example, industrial organization (for a review, see Brandts and Potters 2018) or finance (e.g., Goswami, Noe, and Rebello 1996; Armantier, Holt, and Plott 2013). Further, the lab is increasingly used to study central bank policies (Amano, Kryvtsov, and Petersen 2014; Cornand and Heinemann 2014). Finally, the lab has proved particularly useful to study phenomena that (like stigma) are difficult to observe in practice, such as the *winner’s curse* (Kagel 2003), collusion (Sade, Schnitzlein, and Zender 2006), or corruption (Armantier and Boly 2013).

DW borrowing is less costly than the alternative, but it may be detected by the investor who must decide whether or not to fund the bank. Because DW borrowing is informative about solvency, the investor may decide not to fund a detected bank. This may produce DW stigma as the bank may prefer to choose the costly alternative in order to remain undetected and improve its chances of being funded. An original feature of the model is that, consistent with policy maker's views, the detection probability decreases with the number of DW borrowers. This creates a coordination problem for banks and two pure strategy equilibria: a stigma-free equilibrium where every illiquid bank borrows at the DW and a stigma equilibrium where the DW is not accessed.

In addition to a control treatment, we conduct three main treatments aimed at evaluating policies that have been proposed to mitigate DW stigma. The first treatment explores how DW borrowing responds to lowering the cost of the DW. Starting with Bagehot (1873), the appropriate price of emergency lending has long been debated: making the DW too affordable creates moral hazard, while highly penal rates stigmatize the DW by making it attractive only to the most distressed institutions. In principle, lowering the DW rate should promote DW borrowing and thus reduce stigma. This measure was recommended by Winters (2012) to the BoE to overcome stigma and it was actually implemented by the Fed at the onset of the 2007 crisis (see Internet Appendix 1).

The second treatment evaluates the effect of changing the probability of detection. Clearly, there cannot be stigma unless there is a chance that DW borrowing is detected. As discussed in Internet Appendix 1, concerns have been expressed recently over proposed changes in disclosure policies that could make DW identification easier, thereby exacerbating stigma (e.g., Bernanke 2015a).

The third treatment focuses on a measure proposed by Winters (2012, p. 10) to reduce DW stigma at the BoE: "the Bank should further consider concrete action to reduce any remaining reluctance of banks to use the DW. The best way to accomplish this would be to regularize its use so that crisis usage is less visible and, hence, less stigmatized Drawings would be required in large enough size so that usage in the event of a significant 'genuine' need by a bank would be less likely to be noticed." Thus, the idea is to make DW borrowing both unremarkable and uninformative by making sure banks access the DW regularly, even in normal times when they do not need emergency liquidity. The second treatment tests the effectiveness of one approach that ensures such regular activity: randomly asking banks to borrow at the DW.

The results from the control treatment show that subjects coordinate systematically on the stigma equilibrium. Thus, we are able to generate DW stigma, a phenomenon rarely observed but widely believed to occur in the field. With respect to our main research question, we find that DW stigma is not significantly reduced by cutting the cost of the DW or by making DW borrowing difficult to detect. In contrast, making DW borrowing regular with random borrowing solves the stigma problem in our experiment as it helps subjects coordinate on the stigma-free equilibrium. A series of follow-up experiments

indicates that these results are robust and that random borrowing is only needed temporarily until regular DW usage becomes routine.

1. Related Literature

In Greek language, stigma refers to bodily marks left on persons to be avoided (e.g., slaves, criminals, sinners). Starting with Goffman (1963), the concept of stigma has been applied in the social sciences to a large number of conditions including various mental and physical disorders, sexual orientation, criminal convictions, long-term unemployment, and poverty. As noted by Durlauf and Blume (2008), however, the definition of stigma in this literature is often ambiguous and overlaps with other concepts, such as shame, discrimination, stereotyping, or reputation concerns.

In economics, after early formalizations in specific environments (e.g., Besley and Coate 1992; Rasmusen 1996), general models of stigma have been proposed, notably the “honor-stigma” model of Bénabou and Tirole (2006). In these models, stigma is a form of statistical discrimination triggered by a behavior or characteristic that deviates from the social norm. An essential feature of these stigma models is that norms are formed endogenously. This allows for multiple equilibria, including a corner solution where stigma has a disciplining effect and deviant behavior is nonexistent, and a stigma-free equilibrium where the associated behavior becomes the norm.

Economists distinguish two forms of stigma: exogenous stigma (related, e.g., to growing poor) and participation stigma (related, e.g., to social programs). When it concerns undesirable activities, such as crime (Rasmusen 1996) or tax evasion (Kim 2003), stigma can be a cost-effective alternative to formal punishment. When it concerns socially desirable programs, participation stigma is often welfare reducing. For instance, stigma has been shown to impede the take-up rates in human immunodeficiency virus (HIV) testing (Thorton 2008), mental health treatment (Bharadwaj, Pai, and Suziedelyte 2017), and welfare programs (Moffitt 1983). Participation stigma is also a major concern for emergency lending facilities, government bailouts, and International Monetary Fund (IMF) programs (Ito 2012). In each case, eligible distressed firms or countries refrain from seeking the help they need for fear that it may signal weakness and inhibit future borrowing. Mitigating such participation stigma is thus an unequivocal policy objective.³

Three types of antistigma interventions have been tested in the social sciences using lab and field experiments: *information provision* about the negative consequences of stigmatization and the absence of merit of stereotypes; *perspective taking*, under which one is asked to step into the shoes of a

³ For instance, the IMF stated in 2017 that a key objective of its lending reform was “to reduce the perceived stigma of borrowing from the IMF” (<https://www.imf.org/external/np/exr/fac/faqs.htm#q11>).

stigmatized person; and *personal contact*, under which one interacts with a stigmatized person. Overall, contact interventions have been found to be the most effective (for reviews, see Stuart, Arboleda-Florez, and Sartorius 2012 or Gaebel, Rössler and Sartorius 2017). Bénabou and Tirole (2006) suggest two additional approaches to shift norms and mitigate stigma: making participation difficult to detect (e.g., food stamps are now dispensed with regular-looking credit cards) and having nonconcerned parties take part in the program (e.g., Atkinson [2015] argues that by virtue of being unconditional, universal “basic income” or “free-lunch” programs are less conducive to stigma than classic welfare assistance).

So far, the experiments conducted in economics have focused on measuring participation stigma either in the lab (e.g., Hoffmann, Fooks, and Messer 2014; Friedrichsen, König, and Schmacker 2018) or in the field (e.g., Eriksson and Rooth 2014; Bhargava and Manoli 2015). Lab experiments have also measured “technical stigma” toward risky products, such as genetically modified food (Schulze and Wansink 2012). Our paper expands on this experimental literature by testing specific antistigma policies.

An extensive literature studies LoLR policies (for a review, see Freixas, Rochet, and Parigi 2004). The issue of stigma associated with these programs has been studied both empirically and theoretically. Evidence of DW stigma has been found during three distinct periods: the great depression (Anbil 2018; Vossmeier 2019; Anbil and Vossmeier 2019), the *old* (pre-2003) DW regime (Peristiani 1998; Furfine 2001), and the great recession (Armantier et al. 2015). On the theory side, stigma has been rationalized for different programs: Ennis and Weinberg (2013) and Ennis (2019) focus on DW stigma; La’O (2014), Gauthier et al. (2015), and Hu and Zhang (2019) consider the interaction between the DW and the term auction facility; and Che, Choe, and Rhee (2018) and Gorton and Ordóñez (2019) study government interventions during financial crises. These adverse selection models share the same basic features (e.g., emergency lending is informative about the borrower’s unknown quality) and the same objective (to understand how stigma can emerge in equilibrium). Our model differs in two ways: first, it integrates endogenous social norm, a feature essential in the broader stigma literature (Bénabou and Tirole 2006); second, we focus on antistigma policies.

The literature on LoLR stigma is nevertheless informative for our study. In particular, Anbil (2018) and Vossmeier (2019) find evidence that full transparency can foster stigma, whereas Che, Choe, and Rhee (2018) and Gorton and Ordóñez (2019) argue that bailouts must be covert to be effective. Although we do not consider these corner solutions, we explore how stigma concerns respond to increased DW opacity. Further, in the models of Li, Milne, and Qiu (2016), Che, Choe, and Rhee (2018), and Ennis (2019), firms only accept a bailout when its terms are extremely generous. Thus, small cuts to the

DW rate may have no effect in our experiment. Finally, Anbil (2018) emphasizes the role of banks' coordination in mitigating stigma. By making DW access more regular, the random borrowing policy we propose may help facilitate such coordination. More generally, because we can manipulate DW policies exogenously in the lab, our study expands the LoLR literature by providing new causal evidence about the possible effectiveness of different antistigma measures.

2. The Model

To capture the essence of emergency lending and DW stigma, we develop a coordination game with adverse selection. The objective is not to capture every feature of actual funding markets or to characterize all possible equilibria. Instead, our goal is to develop a model sufficiently rich to produce DW stigma in equilibrium, but simple enough to be studied in the lab.

2.1 Setting

There are two populations of players, banks and investors, each of size N . A bank is matched randomly with an investor. The players move sequentially: the bank moves in period 1, the investor in period 2. Players are risk neutral and do not discount the future. We focus on pure strategies.

The bank can be solvent or insolvent, liquid, or illiquid. There is asymmetric information about the bank's type: it is known to the bank at the beginning of the game, but it is revealed to the investor only at the end of the game. To simplify, we assume that every insolvent bank is illiquid. Thus, there are three types of banks: solvent-liquid with probability P_{SL} , solvent-illiquid with probability $P_{S\bar{L}}$ and insolvent-illiquid with probability $P_{\bar{S}\bar{L}}$. The banks' types are iid and private, but $\{P_{SL}, P_{S\bar{L}}, P_{\bar{S}\bar{L}}\}$ is common knowledge. If the bank is illiquid, then it must borrow 1 unit of liquidity in period 1 either at the DW at a cost dw or from an alternative source at a cost $c > dw$.

Note that we are making five assumptions. First, liquidity is measured at the end of the day and is net of what a bank can obtain in the interbank market. This is in contrast with models of banks' funding in which liquidity shocks are received early in the day and can be addressed first in the interbank market. In our model of *emergency* funding, end-of-day liquidity shocks can only be addressed at the DW or with a costly alternative. Second, because frictions impair the interbank market, both solvent and insolvent banks can have a need for emergency funding. Third, the DW lends to any bank. In practice, the DW should not lend to insolvent banks, but solvency screening is difficult, especially in times of crises (Freixas, Rochet, and Parigi 2004). Here, we assume it is impossible, consistent with, for example, Ennis and Weinberg

(2013).⁴ Fourth, the cost of the alternative c is higher than the DW rate as it captures both monetary (e.g., losses from fire sales) and nonmonetary (e.g., loss of reputation when failing to (re)pay a counterparty) elements. Fifth, a bank returns any unit of liquidity borrowed in period 1 regardless of its solvency status.

In period 2, the investor decides whether or not to take a stake in the bank. To simplify, we say that the investor either funds or does not fund the bank. When the investor funds the bank, he gets V if the bank is solvent or 0 if the bank is insolvent. When the investor does not fund the bank, he gets an outside option v . The bank's operating revenue in period 1 is k . In period 2, the bank's revenue is either k if it is funded by the investor or 0 otherwise as it effectively ceases to operate. Thus, the bank's revenue in both periods does not directly depend on its type. Relaxing this assumption would enrich the model, but it is not necessary for stigma to emerge in equilibrium.

In principle, DW borrowing should remain anonymous, in which case there cannot be stigma. In practice, however, immediate public identification has occurred (see Internet Appendix 1). Further, policy makers believe that real-time detection is more difficult when more banks borrow at the DW simultaneously (Bernanke 2009; Anbil 2018). To capture this "safety in numbers" effect, we assume that $\theta(n)$, the probability that the investor detects the bank's DW borrowing between the two periods when a total of n banks borrow at the DW in period 1, is decreasing in n : $\theta'(n) < 0$. The endogenous nature of the detection probability introduces a coordination component to the game consistent with the broader stigma theory with endogenous norm (Bénabou and Tirole 2006).

As can be seen in the box below where the structure of the game is summarized for reference, observing the bank at the DW enables the investor to update his beliefs about the bank's solvency before making a funding decision. Indeed, DW detection implies that the bank needed emergency funding, which is more likely for insolvent banks. Thus, there is an indirect cost to DW borrowing for the bank: if detected, the investor may decide not to fund the bank because of the increased solvency concern.⁵ In the remainder, we say there is DW stigma when no bank borrows at the DW, and that DW borrowers are stigmatized when investors fund only nondetected banks.

⁴ This assumption is not necessary. DW loans could be refused to insolvent banks with a probability. If this probability gets too close to 1, however, then the opposite of stigma occurs: solvent banks have an incentive to borrow at the DW and to disclose publicly this "certificate of solvency" delivered by the central bank.

⁵ Consistent with, for example, Ennis and Weinberg (2013), the alternative to the DW is unobservable to the investor in the model. This assumption is not necessary to have stigma in equilibrium: DW borrowing and its alternative both could be detected. However, there cannot be DW stigma unless DW detection is a stronger signal of financial weakness.

Structure of the Game

- Period 1:** Each of N banks earns k and receives a private type (solvent-liquid, solvent-illiquid or insolvent-illiquid) independently drawn from a known distribution. Illiquid banks must borrow either at the DW at a cost dw or from another source at a cost $c > dw$. Banks that borrow at the DW are detected with probability $\theta(n)$, where $n \leq N$ is the number of DW borrowers in period 1.
- Period 2:** The investor first learns whether the bank he is matched with was detected at the DW. Then the investor decides whether to fund the bank. If the investor funds the bank, then the bank earns k , whereas the investor earns either V if the bank is solvent or 0 if the bank is insolvent. If the investor does not fund the bank, then the investor earns v , and the bank earns 0 in period 2.

For the model to be sensible, and because we want to design an experiment in which DW stigma can emerge in equilibrium, we focus in the remainder on the following set of parameters:

- A1: $0 < dw < c < k \leq v < V$,
- A2: $\underline{P} < \frac{v}{V} < p$, where $\underline{P} = \frac{P_{SL}}{P_{SL} + P_{\bar{S}\bar{L}}}$ and $p = 1 - P_{\bar{S}\bar{L}}$.

A1 guarantees that (a) DW borrowing is not strictly dominated ($dw < c$), (b) it is profitable for a bank to operate in period 2 even after incurring the higher illiquidity cost c ($k > c$), and (c) it is socially inefficient for the investor to fund an insolvent bank ($k \leq v$).⁶ Because $\underline{P}$ is the probability that a bank is solvent conditional on being illiquid, the first inequality in A2 implies that the investor should not fund a bank that needs emergency funding ($v > \underline{P}V$). Because p is the unconditional probability that a bank is solvent, the second inequality in A2 implies that it is beneficial for the investor to fund a random bank. Thus, A2 provides a potential role for the DW as a screening device for investors: if DW detection reveals that the bank is more likely to be insolvent, then the investor may decide to fund only nondetected banks.⁷

2.2 Equilibrium with fixed detection probability

We define two pairs of strategy profiles, the stigma strategy $St = \{\text{No bank borrows at the DW; investors fund only nondetected banks}\}$, and the no-stigma strategy $NoSt = \{\text{Illiquid banks borrow at the DW; investors fund only}$

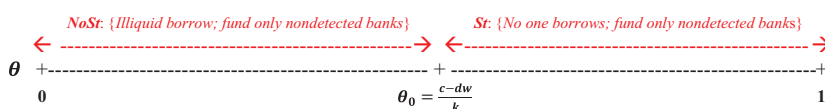
⁶ Under A1, the total surplus when an insolvent bank is not funded (in which case the bank gets 0 and the investor v) is larger than the total surplus when an insolvent bank is funded (in which case the bank gets k and the investor 0).

⁷ The second inequality in A2 also implies that without a DW, the market is not frozen in period 2 because the investor is willing to fund a random bank. In contrast, Che, Choe, and Rhee (2018) consider government bailouts aimed at *jumpstarting* frozen markets. The DW, however, is a *permanent* facility and its primary role is not to jumpstart markets. Finally, it is easy to show that stigma cannot be an equilibrium in our model if we assume that the market is frozen absent a DW.

nondetected banks}. We consider first the case of a fixed detection probability $\theta(n)=\theta$ for Proposition 1 (all the proofs can be found in Internet Appendix 2).

Proposition 1. When $\theta > \theta_0 = \frac{c-dw}{k}$ the unique equilibrium is *St*, otherwise the unique equilibrium is *NoSt*.

As illustrated below, Proposition 1 states that there is a threshold probability θ_0 that makes the bank indifferent between DW borrowing and the alternative. When the probability of detection θ is lower than θ_0 it is a best response for illiquid banks to use the DW, and when θ is higher than θ_0 banks are better off not using the DW. In the extreme, DW borrowing is covert and thus stigma-free when $\theta=0$, while the DW is fully transparent and stigma is inevitable when $\theta=1$.



2.3 Equilibrium with endogenous detection probability

We now return to the case of an endogenous probability of detection $\theta(n)$. We denote the threshold n_0 such that $\theta(n_0)=\theta_0$ and the expected number of illiquid banks $N_{\bar{L}}=(P_{S\bar{L}}+P_{\bar{S}\bar{L}})N$.

Corollary 1. If $1 < n_0 < N_{\bar{L}}$, then there are two equilibria, *NoSt* and *St*. Otherwise, *NoSt* is the unique equilibrium when $n_0 < 1$ and *St* is the unique equilibrium when $n_0 > N_{\bar{L}}$.

Intuitively, Corollary 1 states that if a critical mass of illiquid banks $\bar{n}$ (with $n_0 < \bar{n} < N_{\bar{L}}$) borrow at the DW, then the detection probability $\theta(\bar{n})$ becomes sufficiently low (i.e., $\theta(\bar{n}) < \theta_0$) to make DW borrowing a best response for an illiquid bank. Thus, banks coordinate on DW borrowing in that case and *NoSt* is an equilibrium. Conversely, if the number of DW borrowers $\bar{n}$ is below the threshold (i.e., $1 < \bar{n} < n_0$), then the detection probability $\theta(\bar{n})$ is too high (i.e., $\theta(\bar{n}) > \theta_0$) to make DW borrowing a best response for an illiquid bank. Thus, banks coordinate on not borrowing at the DW in that case and *St* is also an equilibrium.

2.4 Comparative statics

In the remainder, we focus on the most interesting case with multiple equilibria (i.e., when $1 < n_0 < N_{\bar{L}}$). We start by showing that a stigma-free DW is welfare improving.

Proposition 2. Total surplus is higher under *NoSt* than under *St*.

Now our objective is to understand how banks can be nudged to coordinate on the good equilibrium *NoSt*. Of particular interest is dw , the only policy variable fully under the LoLR's control. Because $\theta(n)$ is decreasing in n and $\theta_0 = (c - dw)/k$ is decreasing in dw , we have $\frac{\partial n_0}{\partial dw} > 0$. Thus, reducing the DW cost makes coordination on the stigma-free equilibrium easier in the sense that it lowers the threshold n_0 of DW borrowers required to make *NoSt* a best response. In the extreme, setting $dw = c$ would necessarily create stigma (since $\theta_0 = 0$ and any positive detection probability would produce stigma), while setting $dw = 0$ would not necessarily eliminate stigma (since $0 < \theta_0 = c/k < 1$). Thus, lowering the DW rate makes stigma less likely, but ruling out stigma entirely as a possible equilibrium would require a subsidized (i.e., negative) rate, consistent with Ennis (2019).

We also explore how changing the detection probability from $\theta(n)$ to $\tilde{\theta}(n)$ can affect behavior. If $\tilde{\theta}(n) \leq \theta(n) \forall n$, then $n'_0 = \tilde{\theta}^{-1}(\theta_0) \leq n_0$. Thus, a lower detection probability makes coordination on the stigma-free equilibrium easier in the sense that fewer DW borrowers n'_0 are required to make *NoSt* a best response. This is consistent with the views of policy makers who argue that a more opaque DW is less prone to stigma (see Internet Appendix 1).

2.5 Model extension: Regular random DW borrowing

We now consider one approach to ensure regular DW activity even in normal times: random borrowing. Before a bank learns its type in period 1, it now faces a probability α of being selected to borrow at the DW. The illiquid banks not selected randomly must then decide whether to borrow voluntarily at the DW. Thus, there are now two kinds of DW borrowers: voluntary borrowers (who are necessarily illiquid) and involuntary borrowers (who can be solvent or insolvent).

To simplify, we assume that voluntary and involuntary DW borrowers are indistinguishable. They all borrow 1 unit of liquidity at a cost dw and face the same detection probability $\theta(n + \tilde{n})$, where n and $\tilde{n}$ are the number of banks that borrow at the DW voluntarily and involuntarily, respectively. We define the threshold selection probability $\alpha_0 = \frac{v - PV}{V(p - P)}$ and a new strategy profile *NoSt'* = {*Illiquid banks borrow at the DW; investors fund detected and nondetected banks*}.

Proposition 3. *St* is no longer an equilibrium when $\alpha > 0$. The unique equilibrium is *NoSt* when $\alpha < \alpha_0$, and *NoSt'* otherwise.

The first part of proposition 3 states that introducing a chance (even a small chance) of random borrowing eliminates DW stigma as a possible equilibrium. The intuition behind this result is straightforward: assume that no bank voluntarily borrows at the DW. In that case, the only banks that can be detected are involuntary borrowers. Then the investor is willing to fund every detected bank under A2, because involuntary borrowers are selected randomly

from the pool of all banks. As a result, illiquid banks would now be funded regardless of whether or not they are detected. Therefore, they have a strict incentive to deviate and borrow voluntarily at the DW in period 1 because it is less costly than the alternative. Thus, St cannot be an equilibrium.

The second part of proposition 3 states that, while illiquid banks use the DW in equilibrium, the behavior of investors now depends on the selection probability α . If α is sufficiently large (i.e., $\alpha > \alpha_0$), then enough solvent banks are added to the pool of DW borrowers to make it beneficial to fund a detected bank. In this case, the investor funds every bank and $NoSt$ is the equilibrium. In contrast, when $\alpha < \alpha_0$, the investor funds only nondetected banks and $NoSt$ is the equilibrium.

It is easy to show that a reasonable random borrowing policy can improve welfare. Indeed, under random borrowing, illiquid banks (which now coordinate on the good equilibrium) experience gains, while liquid banks (which now occasionally incur the illiquidity cost dw) experience losses. The balance between these gains and losses is positive as long as α remains low.

To sum up, we developed a model testable in the lab with two equilibria, with and without the form of DW stigma described by Bernanke at the onset of the paper: banks, including solvent banks, avoid the DW for fear that they will be perceived as insolvent and stigmatized by investors. The model also suggests policies that may help coordination on the no-stigma strategy: a lower DW cost, a lower detection probability, and regular random borrowing. A natural first step to test the relative effectiveness of these policies is to conduct a controlled lab experiment in which financially motivated human subjects interact in different exogenously varied treatments.

3. Experiment

The experiment consists of 4 main treatments and 5 robustness check treatments. A treatment has 6 sessions, each with 12 subjects (6 banks and 6 investors) and 25 rounds. Roles are fixed and pairs are formed randomly in each round. Pilot experiments were done at the University of Virginia, and the experiment was conducted at the Toulouse School of Economics. The instructions (see Internet Appendix 3) and decision pages were displayed using the Web-based Veconlab interface. Each session lasted about 90 minutes. Earnings averaged around €17 with a minimum of €11.5 and a maximum of €27.4, plus a show-up payment of €5.⁸

3.1 Control treatment

As described in Internet Appendix 3, subjects in the control treatment play in each round the game with endogenous detection probabilities and no random

⁸ For a comparison of subjects' characteristics in each treatment, see Table 6.1 in Internet Appendix 6.

Table 1
Expected profits and total surplus

Treatment	Equilibrium strategy	Illiquid bank's profit	Liquid bank's profit	Investor's profit	Total surplus
Control	<i>NoSt</i>	14.8	50	69.2	101.6
	<i>St</i>	10	50	66.7	96.7
Low DW cost	<i>NoSt</i>	24.8	50	69.2	106.6
	<i>St</i>	10	50	66.7	96.7
Low detection	<i>NoSt</i>	17.1	50	68.8	102.3
	<i>St</i>	10	50	66.7	96.7
Random borrowing	<i>NoSt</i>	16.4	44.1	68.7	99.0

The calculations exclude the bank's guaranteed operating revenue in period 1.

borrowing presented in Section 2. The parameters of the model are set as follows.⁹ The probabilities for the bank's types are $P_{SL}=1/2$, $P_{S\bar{L}}=1/6$ and $P_{\bar{S}\bar{L}}=1/3$. Thus, the unconditional probability that a bank is solvent is $p=2/3$, while the probability that a bank is solvent conditional on being illiquid is only $\underline{p}=1/3$. The bank's value when solvent is $V=100$ and the investor's outside option is $v=50$. It is easy to verify that A2 is satisfied (i.e., $\underline{p} < v/V < p$). The cost of the DW is half as large as the cost of the alternative, with $dw=20$ and $c=40$. The bank's per period operating revenue is $k=50$, which satisfies A1 (i.e., $0 < dw < c < k \leq v < V$).

The endogenous probabilities of detection are set to $\theta(1)=0.75$, $\theta(2)=0.5$, and $\theta(n)=0.25$ for $n \geq 3$. With $N=6$ and a 50% chance of being liquid, the average number of illiquid banks in each round is $N_{\bar{L}}=3$. The threshold probability is $\theta(n_0)=(c-dw)/k=0.4$, so that $2 < n_0 < 3$. Thus, $1 < n_0 < N_{\bar{L}}$, in which case both *NoSt* and *St* are equilibria according to Corollary 1.

The top row of Table 1 reports the predicted expected profits and total surplus in the control treatment. We can see that coordination on the stigma-free equilibrium *NoSt* instead of *St* leaves liquid banks unaffected; it benefits investors slightly; and it increases illiquid banks' expected profits by about 50%. Nevertheless, note that it is a best response for a bank to borrow at the DW only when there are at least two other DW borrowers (i.e., $n_0 > 2$). Thus, despite substantial financial incentives, subjects may find it difficult to coordinate on *NoSt* in the control treatment.

3.2 Low DW cost treatment

As indicated in Table 2, the *Low DW cost* treatment has the same parameter values as the control treatment except for the DW rate which is reduced by half, from 20 to 10. Because the threshold probability is $\theta(n_0)=0.6$, we now have $1 < n_0 < 2$. Thus, it is a best response for a bank to borrow at the DW if there is at least one other DW borrower. Finally, Table 1 shows that the incentive of illiquid banks to coordinate on *NoSt* increases from 14.8 in the

⁹ As is standard in experimental economics, the parameters chosen for the experiment are not intended to be realistic. Instead, we consider simple environments in which treatment effects have the best chance to emerge.

Table 2
Parameters in each treatment

Treatment	Detection probability when n banks borrow voluntarily at DW			DW cost	Random borrowing probability
	$n = 1$	$n = 2$	$n \geq 1$		
Control	0.75	0.50	0.25	20	0
Low DW cost	0.75	0.50	0.25	10	0
Low detection	0.50	0.25	0.25	20	0
Random borrowing	0.50	0.25	0.25	20	1/6

control treatment to 24.8 in the low DW cost treatment. Hence, coordination on *NoSt* should be easier than in the control treatment.

3.3 Low detection treatment

The *Low detection* treatment has the same parameter values as the control treatment, except that the endogenous detection probabilities are now set at $\theta(1)=0.5$, and $\theta(n)=0.25$ for $n \geq 2$, as shown in Table 2. With these lower detection probabilities DW borrowing becomes a best response as long as there is at least one other DW borrower (i.e., $1 < n_0 < 2$).

3.4 Random borrowing treatment

The *Random borrowing* treatment has the same parameter values as the control treatment. The difference is that in each round, 1 of the 6 banks is picked at random and has to borrow at the DW regardless of its type (i.e., liquid or illiquid, solvent or insolvent). Voluntary and involuntary DW borrowers incur the same cost $dw=20$, and they face the same endogenous probability of detection as in the control treatment. However, because there is always one involuntary DW borrower in each round, the detection probabilities actually faced by the n voluntary borrowers are now 0.5 and 0.25 when $n=1$ and $n \geq 2$, respectively. Thus, the random borrowing and low detection treatments have the same detection probabilities for voluntary borrowers (see Table 2). Our implementation of random borrowing therefore ensures that involuntarily borrowing is the only difference between the two treatments. Note also that the random borrowing probability $\alpha = 1/6$ is below the threshold $\alpha_0=0.5$ from Proposition 3, so that *NoSt* is the only equilibrium in this treatment. Finally, Table 1 shows that compared to *St* in the control treatment, random borrowing benefits investors and illiquid banks, while it is detrimental to liquid banks. In aggregate, however, random borrowing is welfare improving as it increases total expected surplus compared to *St*.

3.5 Predictions

Based on the results in Section 2, the following predictions can be tested in the experiment.

- *PI*: In every treatment, investors fund only banks not detected at the DW.

- **P2:** Banks are more likely to coordinate on the no-stigma equilibrium in the low DW cost and low detection probability treatments than in the control treatment.
- **P3:** Banks coordinate on the no-stigma equilibrium in the random borrowing treatment.

3.6 Additional design features common to all treatments

The experiment is conducted with the “strategy method” under which subjects make decisions for all possible contingencies simultaneously. Specifically, before learning its type, the bank decides whether or not to borrow at the DW if solvent-illiquid and if insolvent-illiquid. Similarly, before learning whether the bank was detected at the DW, the investor makes a funding decision if the bank is detected and if the bank is not detected. After all decisions are made, uncertainty (about the bank’s type and DW detection) is resolved and payoffs are calculated using the subjects’ contingent choices. From a game theoretic perspective, the strategy method is equivalent to the “direct-response method” (under which the game is played sequentially with subjects making a single decision) and it yields the same equilibrium behavior. The advantage of the strategy method is that it produces more data since a subject’s full strategy profile is elicited instead of a single decision. This is particularly relevant in this experiment because half of the banks are solvent on average and therefore would end-up making no decision under the direct-response method.

To promote learning, we provide feedback between rounds. In particular, the bank is told what its type was in the previous round, whether it was detected and the investor’s funding decision. Similarly, the investor is told whether the bank was detected, and when the investor funded the bank, whether or not it was solvent. In addition, summary measures about aggregate behavior in the previous round are provided: every subject is told the proportion of banks that chose to borrow at the DW when solvent-illiquid and insolvent-illiquid, as well as the proportion of investors that chose to fund a bank detected and not detected at the DW. Note that this information about past behavior is irrelevant in the model and thus does not affect equilibrium calculations.

Finally, subjects are given experimental instructions with economic context, with terms like “bank,” “investor,” “insolvent,” and “illiquid.” However, we did not refer to the “discount window” because we were concerned it would not be intuitive. Instead, subjects are told they can borrow from “the central bank in its role as the lender of last resort” (see Internet Appendix 3).

We recognize pros and cons in using the strategy method, providing feedback, and using context in the experiment (for a discussion, see Fréchette and Schotter 2015). As noted above, however, the payoff structure of the game and the associated equilibria remain exactly as described in Section 2. Further, because these design features are held constant across treatments, they cannot explain the treatment effects that may emerge in the experiment.

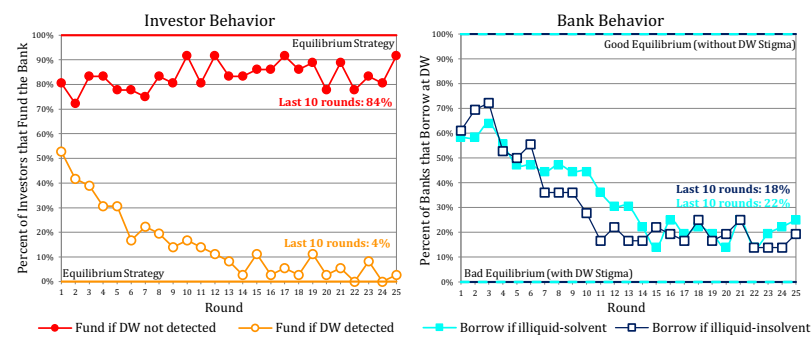


Figure 1
Control treatment
Each line represents the average across the six sessions.

4. Experiment Results

4.1 Control treatment

Figure 1 shows the banks’ and investors’ behavior in the control treatment. The left panel displays the percentage of investors (averaged across the six sessions) that choose to fund banks detected (hollow dots) and not detected (full dots) at the DW in each round. The right panel displays the percentage of banks that choose to borrow at the DW when insolvent-illiquid (hollow squares) and when solvent-illiquid (full squares). Horizontal lines also represent the predicted equilibrium behavior in each panel. Internet Appendix 4 provides figures that display session-level choices.

Starting with the left panel, observe that most investors (81% in round 1) realize immediately that they should fund a bank not detected at the DW. Investors also learn at a rapid pace not to fund a bank detected at the DW: whereas 53% of investors fund a detected bank in round 1, only 17% still do so in round 10. Over the last 10 rounds, investors overwhelmingly stigmatize DW borrowers, as only 4% of detected banks are funded. Overall, investors’ choices in the control treatment become consistent with equilibrium behavior, which provides support for prediction *P1*.

Turning now to the right panel of Figure 1, we first observe that, consistent with theory (since they have the same payoff function), illiquid banks essentially behave in the same way throughout the experiment whether they are solvent or insolvent. Note also that a majority (60%) of banks choose to borrow at the DW in round 1 when illiquid. Further, some evidence suggests that banks initially try to coordinate on the good equilibrium. Indeed, the proportion of DW borrowers increases over the first three rounds. This effort, however, is short lived and quickly unravels. Over the last 10 rounds, stigma is prevalent as only 20% of illiquid banks are willing to borrow at the DW. Thus, the banks’ decisions at the end of the control treatment are generally consistent with the DW stigma equilibrium. Figure 1.1 of Internet Appendix 4 reveals that this

behavior is systematic as coordination on the stigma strategy is observed in all six sessions we conducted.

In summary, behavior in the control treatment is generally consistent with theory. Further, banks' behavior converges toward the stigma strategy in every session we conducted. Thus, we have been able to replicate in the lab DW stigma, a behavior widely believed to occur, but rarely observed in the field. This, in itself, may be considered an interesting result. Next, we turn to the main research question: how can banks' behavior be shifted to the good, stigma-free equilibrium?

4.2 Low DW cost treatment

Figure 2 indicates that, by and large, behavior in the low DW cost treatment is consistent with behavior in the control treatment. Investors almost immediately realize in both treatments that they should fund nondetected banks, and they quickly learn to stigmatize DW borrowers (i.e., not fund detected banks). In fact, we are unable to find a statistical difference in the investors' decisions between the control and the low DW cost treatment at the usual significance levels.¹⁰ Thus, investors' behavior is again generally consistent with theory, which provides additional support for *P1*.

Similar to the control treatment, the majority of banks borrow at the DW in early rounds (64% in round 1), but DW borrowing then declines and stigma becomes prevalent. Average DW borrowing over the last 10 rounds, however, is higher (albeit not significantly) in the low DW cost treatment (32%) than in the control treatment (20%). As indicated by Figure 2.1 of Internet Appendix 4, this difference is driven by a single session, session 5 of the low DW cost treatment, in which subjects were actually able to coordinate on the no-stigma strategy. Without session 5, DW borrowing over the last 10 rounds of the low DW cost treatment drops to 22%, in line with the control treatment average of 20%. Therefore, we find no clear evidence to support prediction *P2*.

Summing up, a 50% cut in the cost of the DW enabled coordination on the stigma-free equilibrium in 1 of the 6 sessions conducted. Overall, however, reducing the DW rate fails to solve the stigma problem: ultimately, most banks in the low DW cost treatment avoid the DW.

Would an even more attractive DW rate prevent stigma? To test this hypothesis, we conduct an additional treatment in which the DW cost is cut once more in half to 5. The results reported in Internet Appendix 5 show that this additional cut still does not solve the stigma problem. Thus, it appears that the subjects' reluctance to approach the DW does not stem from its cost and that reducing the DW rate may not be an effective strategy to address stigma.

¹⁰ Treatment effects are determined by comparing the distributions of session averages (over the last 10 rounds) across treatments using a two-tail permutation test (a nonparametric exact test, Good 2000) of differences in medians.

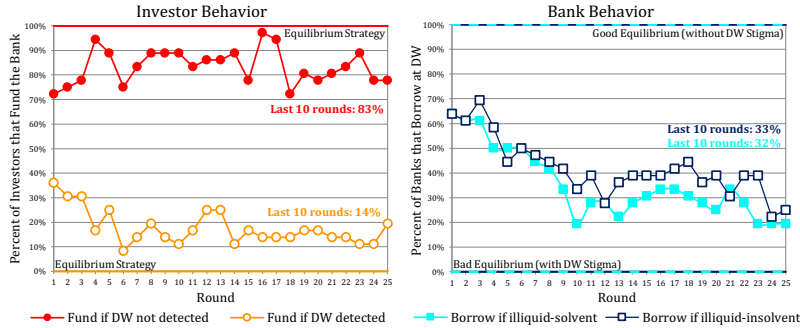


Figure 2
Low DW cost treatment
Each line represents the average across the six sessions.

These results are consistent with those of Ennis (2019), who argues that the DW cannot be effective unless its rate is subsidized (i.e., below market rates). Our results also echo the Fed’s experience. In August 2007, the Fed took a number of measures to stimulate DW borrowing (see Internet Appendix 1). Among them, the DW “penalty” spread over the target rate was reduced by 50%, from 100 basis points to 50, thereby reducing the DW rate from 5.75% to 5.25%. Similar to our experimental results, this policy change was ineffective as DW usage increased little in the second half of 2007.

4.3 Low detection treatment

Figure 3 shows the results in the low detection probability treatment. Overall, the same dynamics of play emerge in the low detection and control treatments. Despite some differences in early rounds, investors learn to fund only nondetected banks in both treatments. Further, after an early attempt at coordinating on the good equilibrium, most banks settle on the DW stigma strategy. In fact, we are unable to find a significant difference (at the usual significance levels) between choices over the last 10 rounds in the control and low detection treatments for banks and investors.

To test whether a more opaque DW remains stigmatized, we conduct a new treatment in which the detection probability is lowered to $\theta(n)=0.25$ regardless of the number of DW borrowers n . The fixed detection probability is now always below the threshold $\theta_0=1/3$ from Section 2, making the no-stigma strategy the unique equilibrium. The results reported in Internet Appendix 5 show that banks do not play this equilibrium. Instead, the proportion of DW borrowers hovers around 50% throughout the experiment. Thus, making the DW more opaque, even beyond the point where DW borrowing becomes the unique equilibrium, does not solve the stigma problem in the experiment.

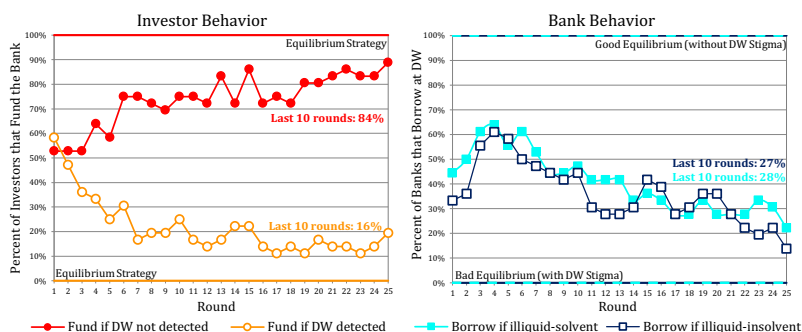


Figure 3
Low detection treatment

Each line represents the average across the six sessions.

4.4 Random borrowing treatment

We now turn to Figure 4, which displays the random borrowing treatment behavior. As in the previous treatments, investors fund banks not detected at the DW. In fact, the funding behavior of investors who face a nondetected bank cannot be distinguished statistically across the four treatments at the usual significance levels. In contrast, funding behavior toward banks detected at the DW is statistically different (at the 5% level) in the random borrowing treatment. While investors learn not to fund a detected bank in the other treatments, we can see in the left panel of Figure 4 that the funding trend is slightly upward in the random borrowing treatment. Nevertheless, the majority of investors still stigmatize DW borrowers (the 50% threshold is not crossed in any round). This treatment effect may be explained by the fact that random borrowing adds solvent-liquid banks to the pool of detected banks, which makes the investors' funding decisions less obvious. Over the last 10 rounds of the control treatment, the expected profit of an investor who funds a detected bank is 29.8, which is clearly dominated by the investor's outside option of 50. In contrast, an investor's expected profit when funding a detected bank in the last 10 rounds of the random borrowing treatment is 44.5, which is closer to his outside option of 50.

Turning now to banks' behavior, we can see in the right panel of Figure 4 a clear and highly significant (at the 1% level) treatment effect. In contrast with the other treatments, banks learn to access the DW voluntarily in the random borrowing treatment. In fact, Figure 4.1 of Internet Appendix 4 reveals that banks systematically learn to coordinate on the stigma-free equilibrium in all six sessions of the random treatment. Thus, consistent with prediction **P3**, we find that subjects converge to the no-stigma equilibrium when DW usage is made regular by random borrowing.

We report in Table 3 average profits and total surplus in the last 10 rounds of each treatment. Observe first that profits in the experiment are well below the

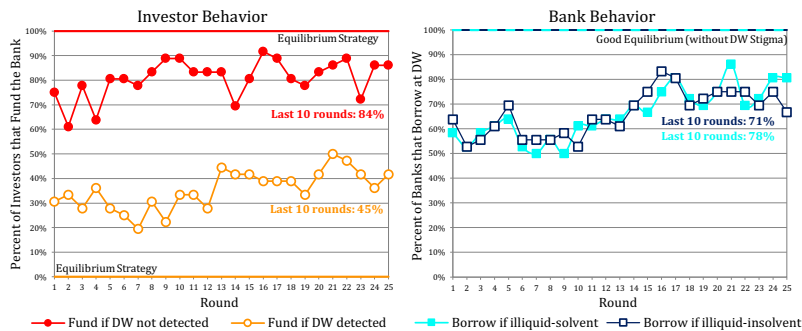


Figure 4
Random borrowing treatment
Each line represents the average across the six sessions.

Table 3
Profits and total surplus in each treatment

Treatment	Illiquid bank's profit	Liquid bank's profit	Investor's profit	Total surplus
Control	1.66 (1.97)	43.14 (2.89)	63.47 (7.91)	85.06 (12.35)
Low DW cost	6.38 (6.53)	39.97 (4.58)	61.94 (4.93)	84.72 (8.18)
Low detection probability	2.68 (6.12)	40.86 (2.02)	61.39 (2.82)	82.56 (6.61)
Random borrowing	14.12** (2.87)	36.65 (4.95)	61.94 (5.07)	86.94 (4.96)

Calculated over the last 10 rounds, excluding the bank's guaranteed operating revenue in period 1. Numbers in parentheses refer to standard deviations. ** $p < .05$.

theoretic benchmarks in Table 1. Nevertheless, Table 3 shows that total surplus is highest in the random borrowing treatment. Further, the magnitude of the increase compared to the control treatment (+2.1% in Table 3) is consistent with theory (+2.4% in Table 1). This treatment effect, however, is too small to be significant. Table 3 also shows that the average profit of illiquid banks is significantly higher (at the 5% level) in the random borrowing treatment. This treatment effect is consistent with theory because illiquid banks should benefit the most from random borrowing as discussed in Section 2.

4.5 Coordination channels

We have just established that, unlike cutting the DW rate or lowering the detection probability, random borrowing was effective at helping subjects coordinate on the stigma-free equilibrium. Three channels could explain this result: with random borrowing, voluntary borrowers face lower detection probabilities (*detection*); involuntary borrowing teaches banks the value of DW

borrowing (*experience*); and investors are more likely to fund detected banks (*funding*).

We can test the first channel by comparing the low detection and random borrowing treatments. As explained earlier, the *detection* channel is common to both treatments (i.e., voluntary borrowers face the same lower detection probabilities), whereas the other two channels are muted in the low detection treatment. Because behavior converges toward the no-stigma equilibrium in the random borrowing treatment, but not in the low detection treatment, we can rule out the *detection* channel.

An econometric analysis (see Internet Appendix 6) provides evidence for the other two channels. First, consistent with the *experience* channel, we find that after a bank is randomly selected to borrow and not detected, it is 35% more likely to return to the DW voluntarily in the next round. Second, consistent with the *funding* channel, we find that the banks' willingness to borrow at the DW increases with the number of investors who fund detected banks in the previous round. In the next section, we test experimentally whether the *funding* channel is necessary to solve DW stigma.

5. Robustness Checks

This section reports on additional treatments that were conducted to evaluate the extent to which random borrowing is a robust mechanism to address DW stigma.

5.1 Low random borrowing treatment

We start by testing the effectiveness of random borrowing when the probability of being asked to borrow at the DW is lower. The *Low random borrowing* treatment is the same as the random borrowing treatment with two exceptions. First, the random borrowing probability is cut in half from 1/6 to 1/12. Second, uncertainty about random borrowing is resolved differently. In each round of the random borrowing treatment, there is always one involuntary DW borrower selected at random from the six banks. In contrast, the number of involuntary DW borrowers can vary from 0 to 6 in the low random borrowing treatment as each bank faces a 1 in 12 chance of being selected.

Under Proposition 3, behavior should still converge to the unique no-stigma equilibrium. As shown in Figure 5, the experimental evidence is largely consistent with this prediction. However, banks appear to take longer to coordinate on the no-stigma strategy than in the random borrowing treatment (Figure 4). An econometric analysis reveals that this difference in learning speed is due mostly to the fact that banks have fewer opportunities to learn from involuntary borrowing when they are selected less often to borrow at the DW. Nevertheless, our results suggest that, consistent with theory, low probabilities of involuntary borrowing may be sufficient to mitigate DW stigma.

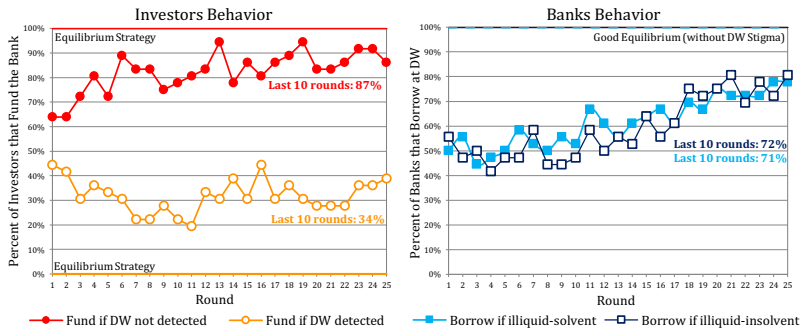


Figure 5
Low random borrowing treatment
Each line represents the average across the six sessions.

5.2 Automaton investors and temporary random borrowing

We now address two questions. First, recall that banks coordinate on DW borrowing in the random borrowing treatment in part because investors deviate from the equilibrium and fund detected banks. So, can banks still coordinate on the good equilibrium without this *funding* channel? Second, Winters’s proposal to address stigma at the DW is “to regularize its use.” Random borrowing is only used as a tool to ensure such regular activity. So, is random borrowing necessary to sustain the no-stigma equilibrium, or is it just a temporary measure that helps normalize DW usage?

To address these questions, we conduct an additional treatment identical to the random borrowing treatment with three exceptions. First, the experiment now consists of 30 rounds instead of 25. Second, banks are told that investors are played by automatons who follow the equilibrium strategy; that is, they invest only in nondetected banks. Third, subjects are informed at the end of round 20 that random borrowing is removed until the end of the experiment. Thus, subjects in the last 10 rounds play exactly the same game as in the control treatment.

The results presented in Figure 6 show that banks still converge to the no-stigma strategy within the first 20 rounds. In fact, convergence may be faster than in the random borrowing treatment (Figure 4). This may be because the strategic environment is arguably simpler in this additional treatment. Indeed, there is no uncertainty about the investor’s behavior, and banks are now playing a pure coordination game. Thus, it appears that the *funding* channel is not necessary to eliminate stigma. Banks can learn to coordinate on the stigma-free equilibrium even when investors systematically stigmatize detected DW borrowers. Instead, we find that the experience acquired from regular involuntarily borrowing may be sufficient for subjects to overcome DW stigma.

Figures 6 shows a clear restart effect after random borrowing is removed at the end of round 20. The effect, however, is short lived, and almost all banks return

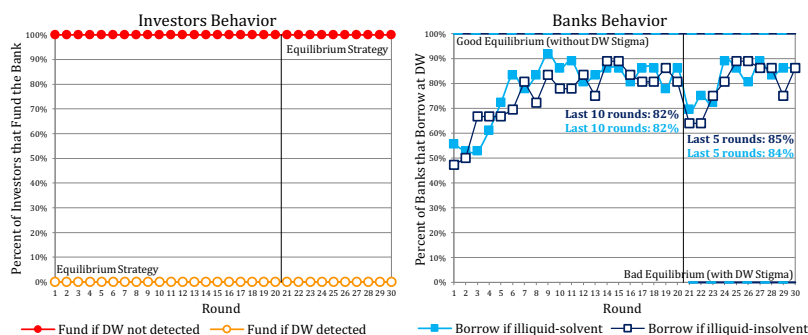


Figure 6
Automaton investors and temporary random borrowing treatment

Each line represents the average across the six sessions. The vertical line represents when random borrowing is removed.

to the DW. Thus, random borrowing is just a means to an end in the experiment, a temporary tool that helps banks coordinate on regular DW usage. Once the habit is formed, it appears sufficient to sustain a stigma-free DW.

5.3 Real-time and delayed detection

As discussed in Internet Appendix 1, the Fed started disclosing the identity of every DW borrower with a 2-year lag in 2010. How does such delayed detection affect stigma concerns and the effectiveness of random borrowing? As a first step in answering this question, we conduct an additional treatment, identical to the random borrowing treatment with one exception. Before the investor makes a funding decision, he is told whether or not the bank he is matched with borrowed at the DW in the previous round. Thus, DW borrowers now face certain delayed detection along with possible real-time detection. In contrast with real-time detection, delayed detection is not directly relevant for the investor because it is not informative about the bank's solvency status in the current round. Nevertheless, stigma may have a lasting effect: with delayed detection a bank may develop a reputation as a DW borrower which may influence the investors funding decision.

We can see in Figure 7 that delayed detection does not seem to affect behavior or the effectiveness of random borrowing: investors learn to stigmatize detected banks and random borrowing helps banks coordinate on the no-stigma equilibrium. This does not imply, however, that investors ignore the new information provided. To see this, we add in the left panel of Figure 7 two lines indicating the percentage of investors who fund a bank when they are told the bank accessed (orange dotted line) or did not access (red dotted line) the DW in the previous round. Figure 7 shows that investors initially use delayed detection to discriminate: in the first 14 rounds, a bank that borrowed at the DW in the previous round is always less likely to be funded. In later rounds, however, there is little funding differences based on delayed detection,

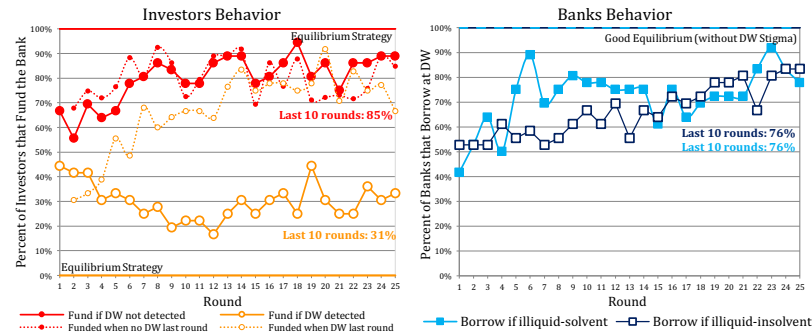


Figure 7
Real time and delayed detection treatment
Each line represents the average across the six sessions.

probably because investors figure out that the signal is not informative about solvency.

6. Conclusion

DW stigma is a threat to financial stability and an impediment to core responsibilities of central banks (i.e., acting as a lender of last resort and implementing monetary policy). Yet, little is known about DW stigma, in part because of the absence of data. In this paper, we propose a coordination game with adverse selection that possesses two equilibria, with and without DW stigma, and we use experimental methods to test policies that have been proposed to overcome DW stigma.

Consistent with behavior believed to occur in the field, stigma prevails in the control treatment. Lowering the cost of the DW is found to have little effect on stigma. This is consistent with anecdotal evidence: in August 2007, the Fed's was unable to stimulate DW borrowing by cutting the DW rate. Similarly, lowering the detection probability fails to address DW stigma in the experiment. This result challenges the view that stigma concerns can be attenuated by making DW borrowing relatively more opaque. Finally, consistent with theory, behavior converges to the stigma-free strategy when the use of the DW is made regular by random borrowing. Additional robustness check treatments confirm that these results are robust. In particular, random borrowing remains effective when the probability of involuntary borrowing is low, when detected banks are automatically not funded, and when past DW visits are disclosed. Further, we find that random borrowing may be needed only temporarily until using the DW becomes the norm. These results therefore support those of Winters (2012), who argues that regular DW usage can mitigate stigma.

We believe this first experimental study of emergency lending contributes new insights about DW stigma and how to overcome it. Our results, however,

should not be overgeneralized. In particular, the effectiveness of random borrowing would have to be confirmed more generally with additional experiments, more comprehensive models and possibly, small scale field experiments. Further, implementing random borrowing would raise several practical challenges. Notably, the DW is currently available on a voluntary basis at the borrower's request. To ask banks to borrow at random times, the Fed would have to embark in substantial modifications of its DW policies, similar to the 2003 reform (see Internet Appendix 1). Nevertheless, the main take-away from this study is that stigma is not inherent to the DW and can be avoided through regular DW activity. Random borrowing is one way to normalize DW borrowing, but there may be more practical alternatives. For instance, routine DW activity could be incentivized by offering better terms to regular DW borrowers. We leave it to future research to identify how best to promote routine DW usage.

It should be noted that the premise for our analysis is that the LoLR responsibilities are assigned to the DW. Our objective was then to explore adjustments to the DW that can alleviate stigma concerns. In other words, we focused on solutions that could be implemented in the relatively short term and without major disruptions. Of course, this should not prevent a more general discussion about DW alternatives and optimal LoLR mechanisms. Such possible alternatives include transforming the DW into a fee-based contingent liquidity facility (Nelson 2017), the "pawnbroker" approach of King (2016b), open market type auctions (Selgin 2017), and regular liquidity auctions (similar to the Fed's Term Auction Facility or the BoE's long-term repo auctions). Some of these alternatives, however, may require an act of Congress to be implemented. Further, as argued by Hauser (2014), these approaches are not perfect substitutes for the DW and may be best described as "lender of *next* to last resort."¹¹ Thus, there still may be a need for a stigma-free DW-like bilateral standing facility where a financial institution can address any idiosyncratic liquidity shock in real time before it festers into a solvency or a systemic problem.

Fixing the DW may not seem like a pressing problem in the current high reserves environment in which the need for liquidity support is unlikely. However, DW stigma is starting to become relevant again as the size of the Fed's balance sheet is reduced and liquidity becomes less abundant. For instance, the DW failed to serve as a safety valve on September 17, 2019, when overnight lending rates suddenly spiked to up to 10% (see Internet Appendix 1). As argued by Bernanke (2015b), one does not fix the fire code once the fire has started. In other words, it may be prudent to address DW stigma *ex ante* in normal times, rather than wait until the onset of the next financial crisis to do so. The value of implementing regular random DW borrowing in normal

¹¹ Because liquidity shocks can occur at any time, the BoE decided to supplement periodic liquidity auctions with a DW facility available on a continuous basis.

times finds additional support in Bernanke (2016), who argues that stigma was less prevalent in the Eurozone during the 2008 financial crisis, in part because European banks routinely engaged with the central bank in normal times. Finally, understanding DW stigma is important for the design of new liquidity facilities. For instance, the “standing repo facility,” recently proposed by Andolfatto and Ihrig (2019) to implement monetary policy, is similar to the DW in most respects (e.g., eligibility, penalty rate, disclosure) except one: it would only accept high quality collateral. However, whether this change is sufficient to make the facility stigma-free and thus effective, is an empirical question.

We conclude by discussing the implications of our results for other forms of stigma in finance and economics (e.g., bailout or welfare stigma). It is interesting to note that the policies that have been experimented to address stigma in some of these cases share the same fundamental principle with the random borrowing approach we propose: having nonconcerned parties take part in the program to dilute any negative signal from participation. In particular, when the Troubled Asset Relief Program (TARP) was launched in 2008, the nine largest U.S. banks were compelled to accept capital injections to promote take up from smaller banks and prevent “TARP stigma.”¹² Similarly, to stimulate DW borrowing, the Fed asked four large banks to visit the DW privately on August 7, 2007. This effort, however, was unsuccessful in part because it was perceived as stage managing (see Internet Appendix 1). Finally, to eliminate the stigma attached to being eligible for free-lunch programs, several school districts have recently adopted policies under which lunch is provided for free to all students regardless of income.¹³ Our results provide support for these approaches, as we find evidence that normalizing participation can overcome stigma.

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¹² Geithner (2014, p. 238) reports “I warned the bankers that if they all didn’t accept the capital, TARP would be stigmatized.”

¹³ To promote health and academic performance, many U.S. public schools have made low-income students eligible to receive free or reduced-price lunch. However, most eligible students (about two-thirds, according to Schwartz and Rothbart 2020) do not take part in the program and forgo lunch because of the stigma associated with their classmates finding out they qualify. Two recent studies suggest that universal free-lunch programs improve academic performance (Schwartz and Rothbart 2020) and students’ behavior (Altindag et al. 2020).

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